EXISTENCE OF THE UNIVERSAL FLATTENING OVER AN ARTINIAN BASE, GIVEN BY A CLOSED IMMERSION, HENCE ALSO A FLATTENING STRATIFICATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The functor of flat quotients of a quasi-coherent module, and its compatibility with an affine open covering. Same chapter flat.tex, the functor F_{flat} discussed in the surrounding section of the chapter (used at proof lines 6575-6577). Over the covering $X = \bigcup U_i$ the functor is the product of the functors of the pairs $(U_i, \mathcal{F}|_{U_i})$, which reduces the claim to the affine case, one closed condition at a time.
- The affine case of the universal flattening over an Artinian base. More on Algebra, label more-algebra-lemma-flattening-artinian-universal-property (invoked at proof line 6582). It supplies the universal flattening in the affine case, realized by a closed subscheme of the Artinian base, which is therefore also a flattening stratification.

Source reference. Original lemma 0525: sources/stacks-chow/tags.txt, source lines 6558–6558. Navigation pointer for \ref{more-algebra-lemma-flattening-artinian-universal-property} used inside the counted span (line 6582); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be the spectrum of an Artinian ring. For any scheme X over S, and any quasi-coherent $\mathcal{O}_X$-module there exists a universal flattening. In fact the universal flattening is given by a closed immersion $S' \to S$, and hence is a flattening stratification for $\mathcal{F}$ as well.*

Proof. Choose an affine open covering $X = \bigcup U_i$. Then F_{flat} is the product of the functors associated to each of the pairs $(U_i, \mathcal{F}|_{U_i})$. Hence it suffices to prove the

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result for each $(U_i, \mathcal{F}|_{U_i})$. In the affine case the lemma follows immediately from More on Algebra, Lemma 0525. $\square$